Plotting directly using pandas

PYTHON FOR R USERS

Daniel Chen
Instructor

Plotting in Python

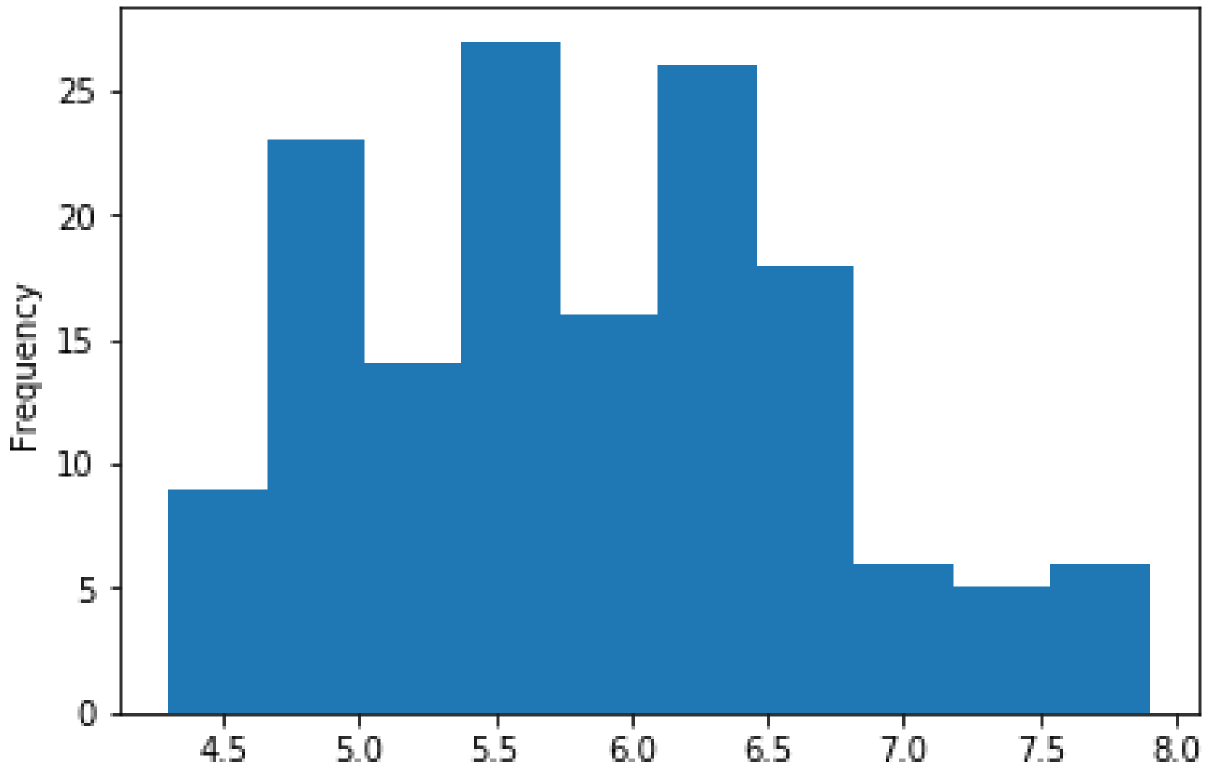
- Quickly show data patterns
- Plotting methods in Python:
  - Pandas
  - Seaborn
  - Matplotlib

Pandas plot method

- `plot()` method
- Works on the pandas DataFrame and Series objects
- Pass `plot` the `kind` argument
- `kind` of plots:
  - `line` : line plot (default)
  - `bar` : vertical bar plot
  - `barh` : horizontal bar plot
- `hist` : histogram
- `box` : boxplot
- `kde` : Kernel Density Estimation plot
- `density` : same as 'kde'
- `area` : area plot
- `pie` : pie plot
- `scatter` : scatter plot
- `hexbin` : hexbin plot

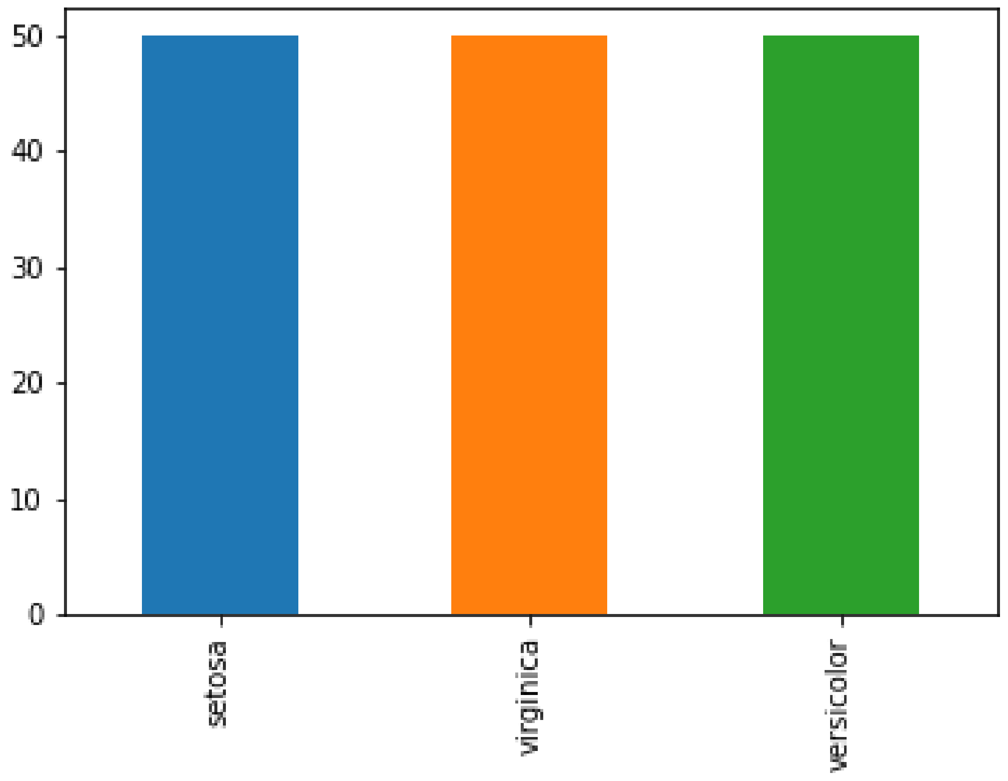
Univariate: Histogram

```
import matplotlib.pyplot as plt
iris['sepal_length'].plot(kind='hist')
plt.show()
```



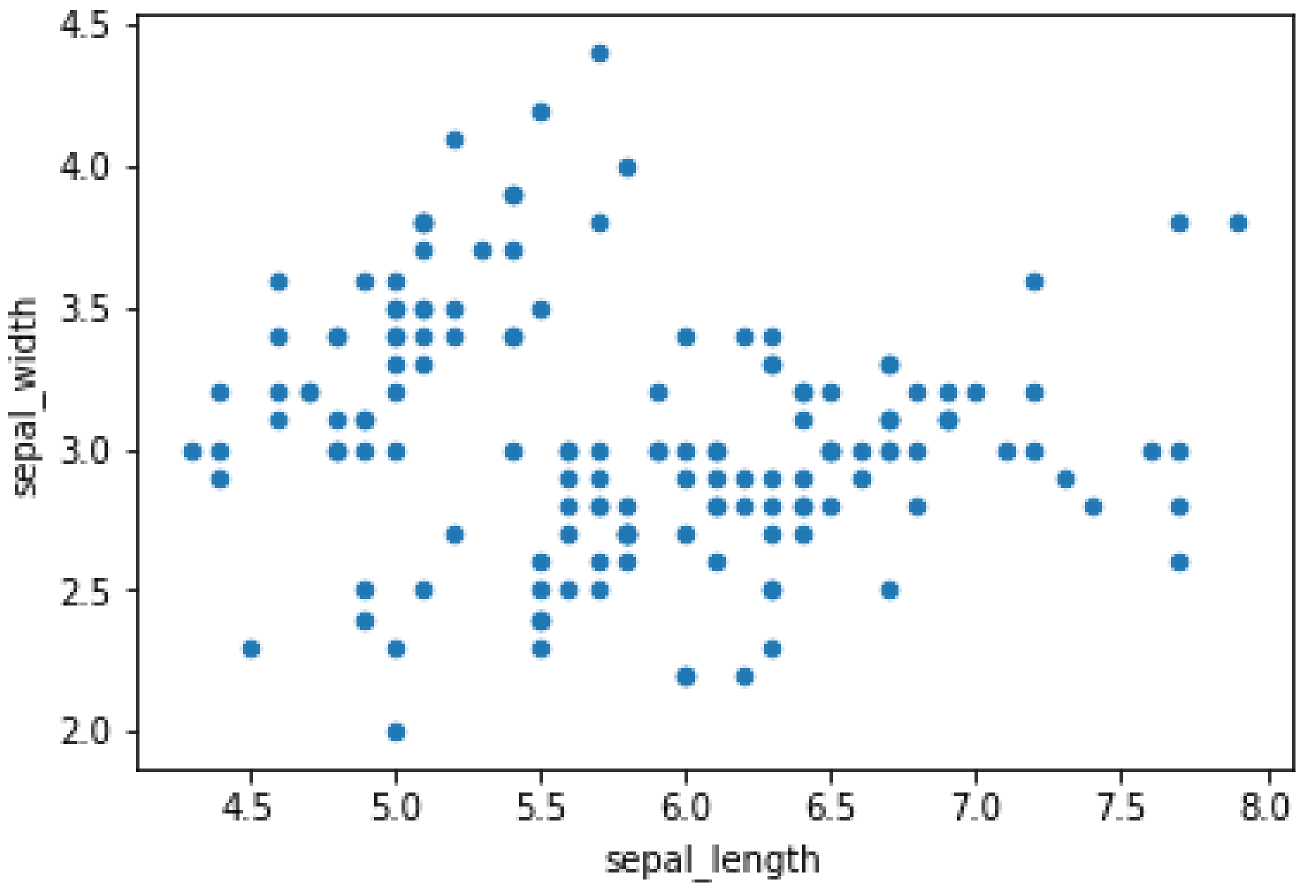
Univariate: Bar plot

```
cts = iris['species'].value_counts()
cts.plot(kind='bar')
plt.show()
```



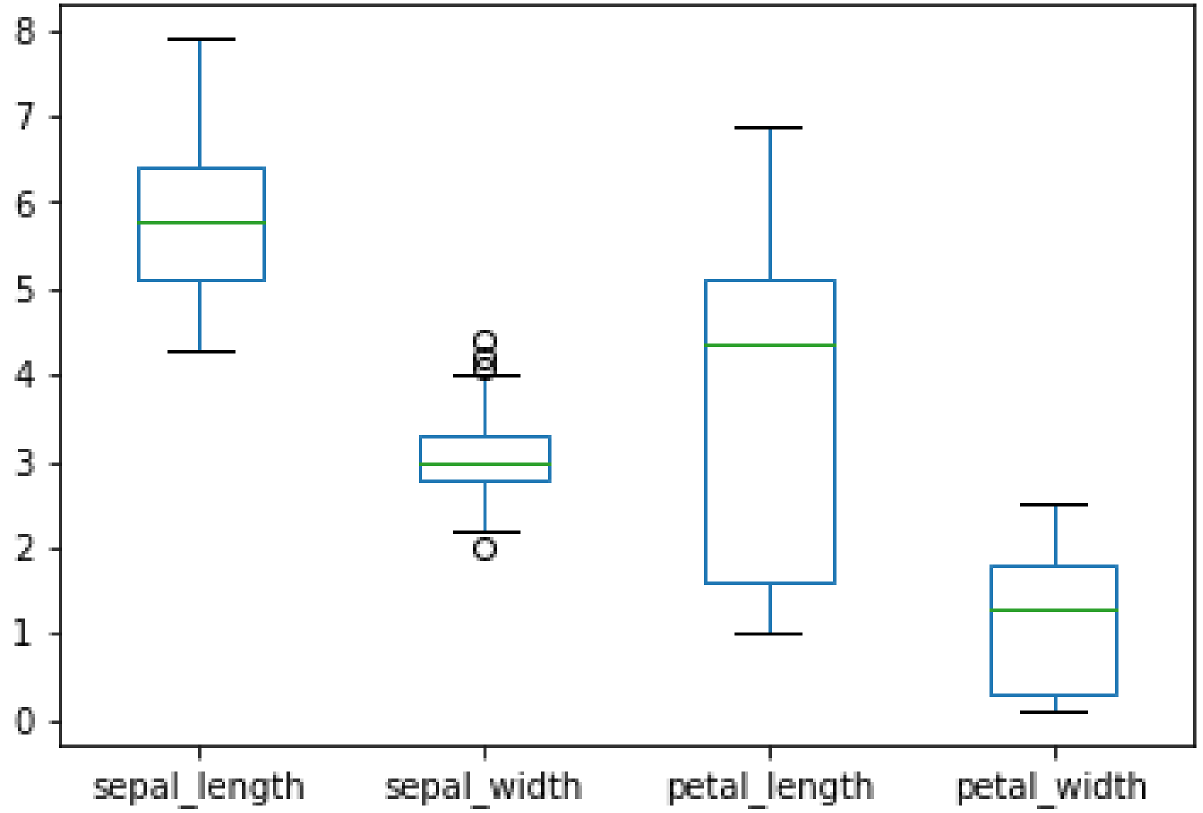
Bivariate: Scatter plot

```
iris.plot(kind='scatter', x='Sepal.Length', y='Sepal.Width')
plt.show()
```



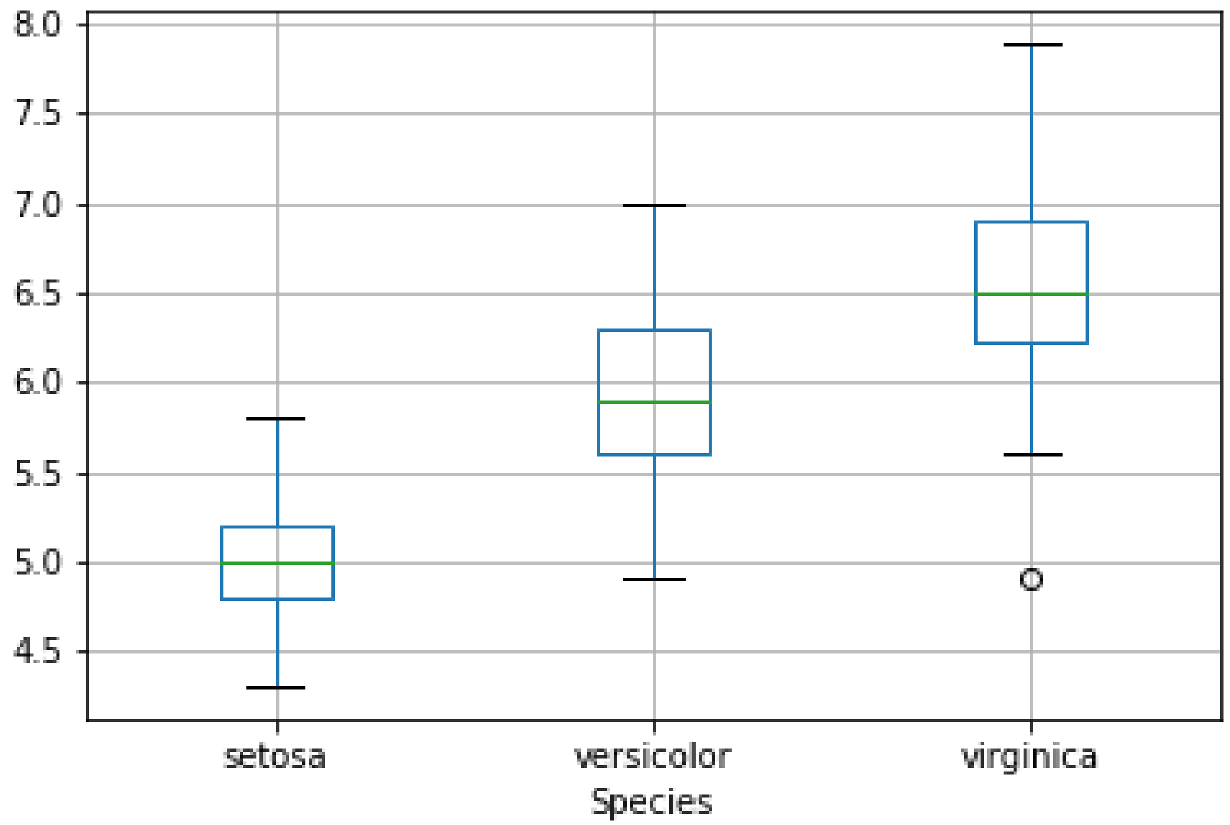
Bivariate: Boxplots

```
iris.plot(kind='box')
plt.show()
```



Bivariate: Boxplots

```
iris.boxplot(by='Species', column='Sepal.Length')
plt.show()
```



Let's practice!

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Seaborn

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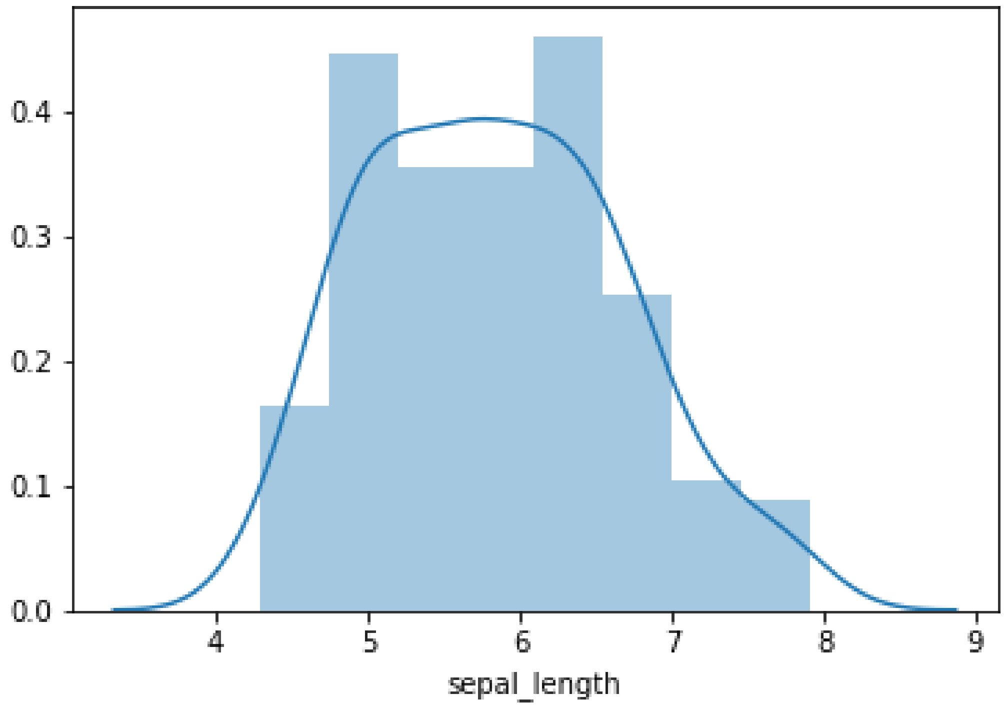
Seaborn

- barplots: `barplot`
- histograms: `displot`
- boxplot: `boxplot`
- scatter plot: `regplot`
- Seaborn benefits
  - colored points by data
  - facet plots by data

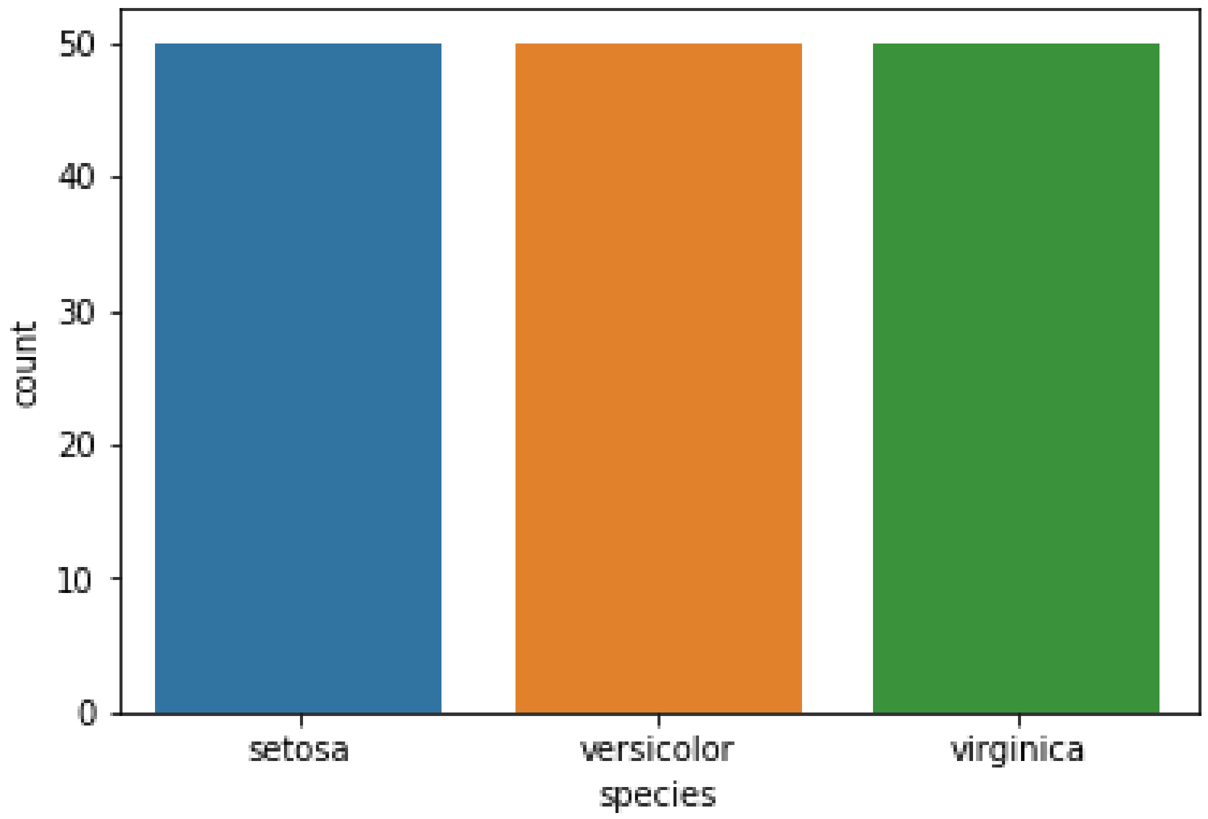
Seaborn histograms

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.distplot(iris['sepal_length'])
plt.show()
```



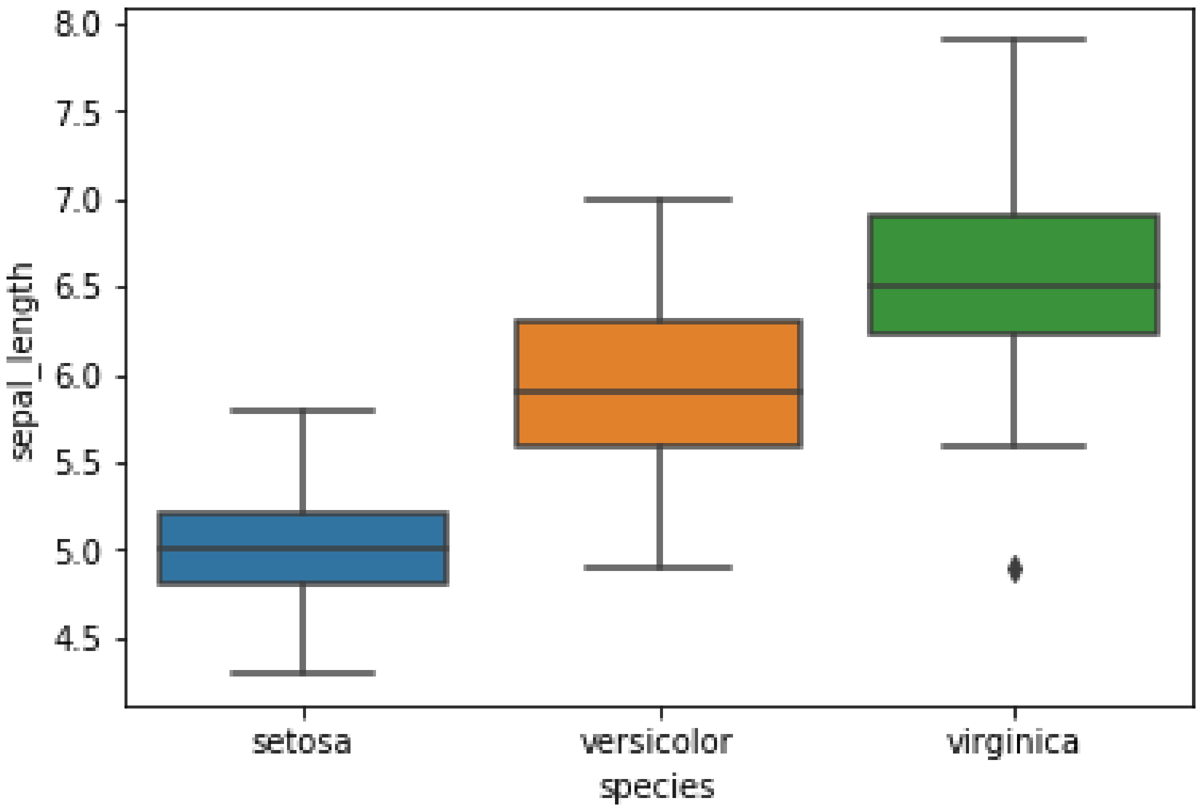
Seaborn count plot

```
sns.countplot('species', data=iris)
plt.show()
```



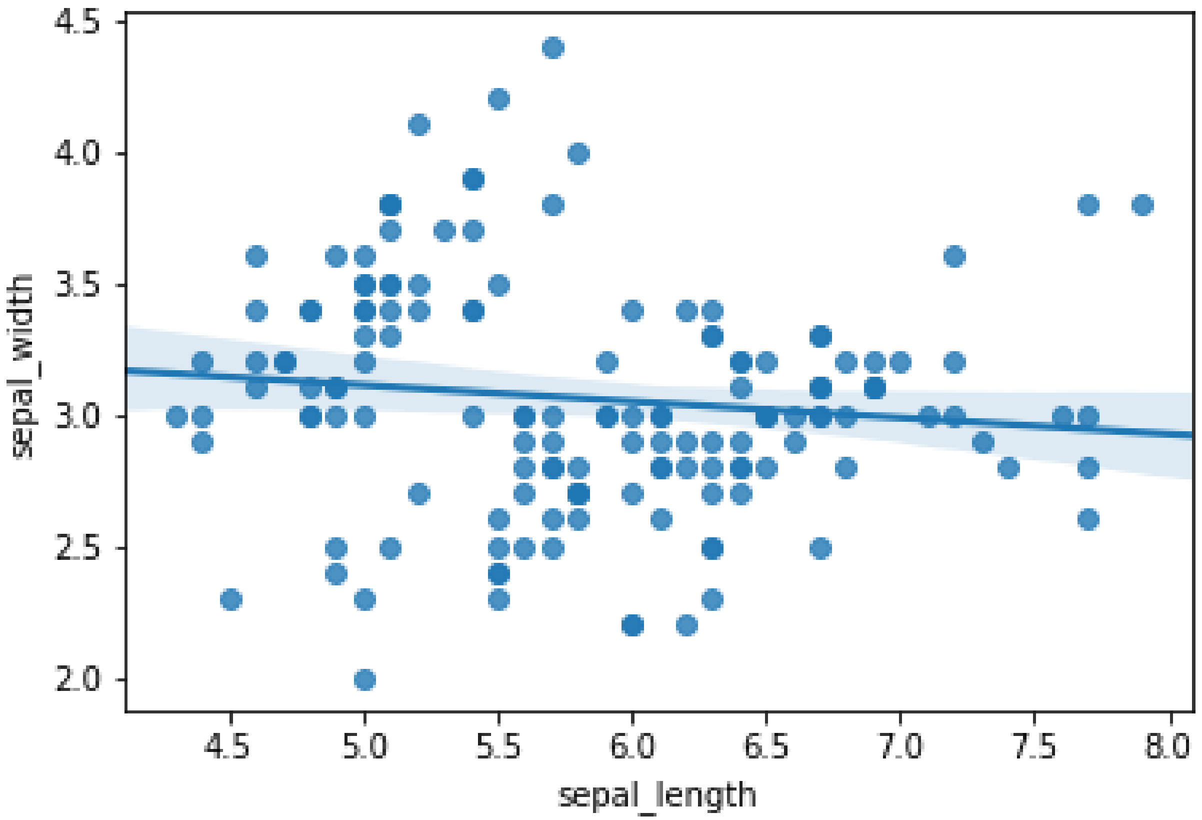
Seaborn boxplots

```
sns.boxplot(x='species', y='sepal_length', data=iris)
plt.show()
```



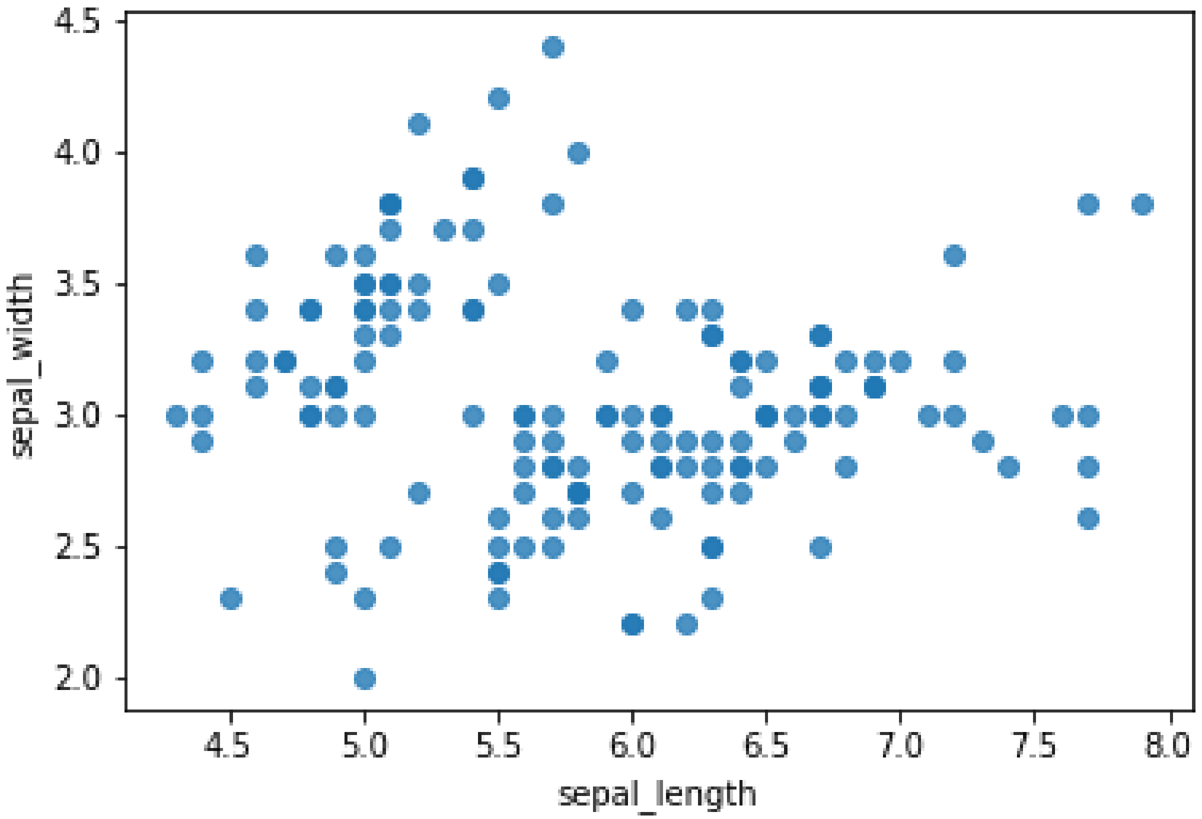
Seaborn scatterplots

```
sns.regplot(x='sepal_length', y='sepal_width', data=iris)
plt.show()
```



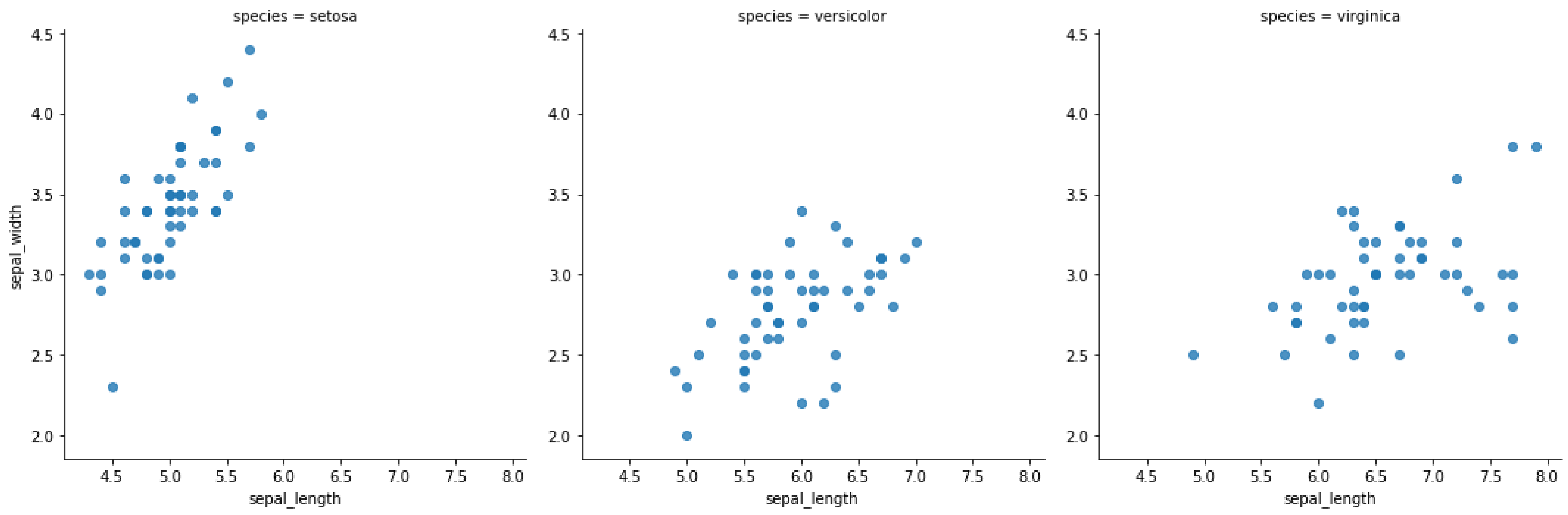
Seaborn scatterplots w/out regression line

```
sns.regplot(x='sepal_length', y='sepal_width', data=iris,
            fit_reg=False)
plt.show()
```



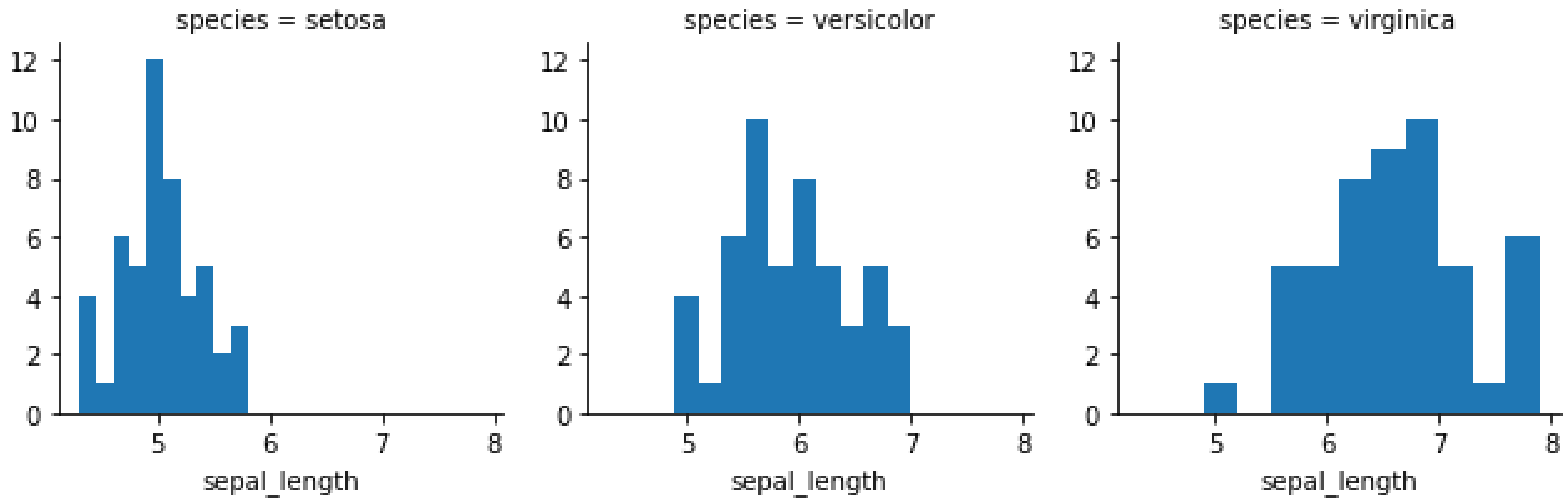
Seaborn facets

```
sns.lmplot(x='sepal_length', y='sepal_width', data=iris,
           fit_reg=False,
           col='species')
plt.show()
```



Seaborn FacetGrid

```
g = sns.FacetGrid(iris, col="species")
g = g.map(plt.hist, "sepal_length")
plt.show()
```



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Matplotlib

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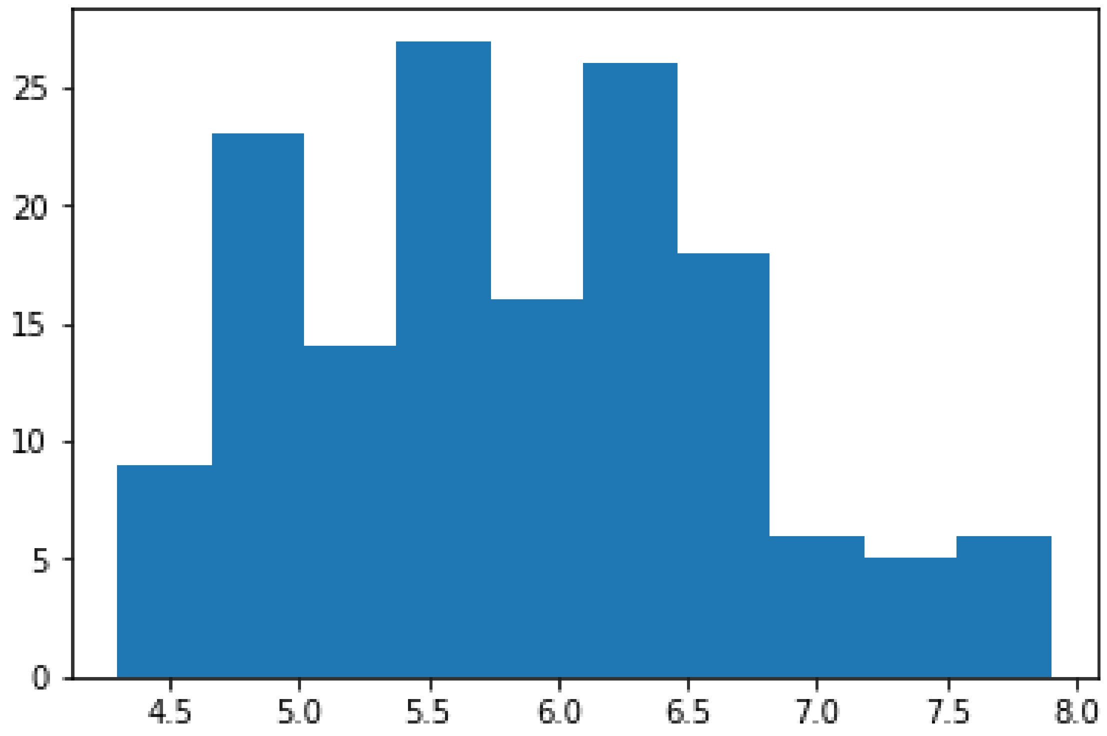
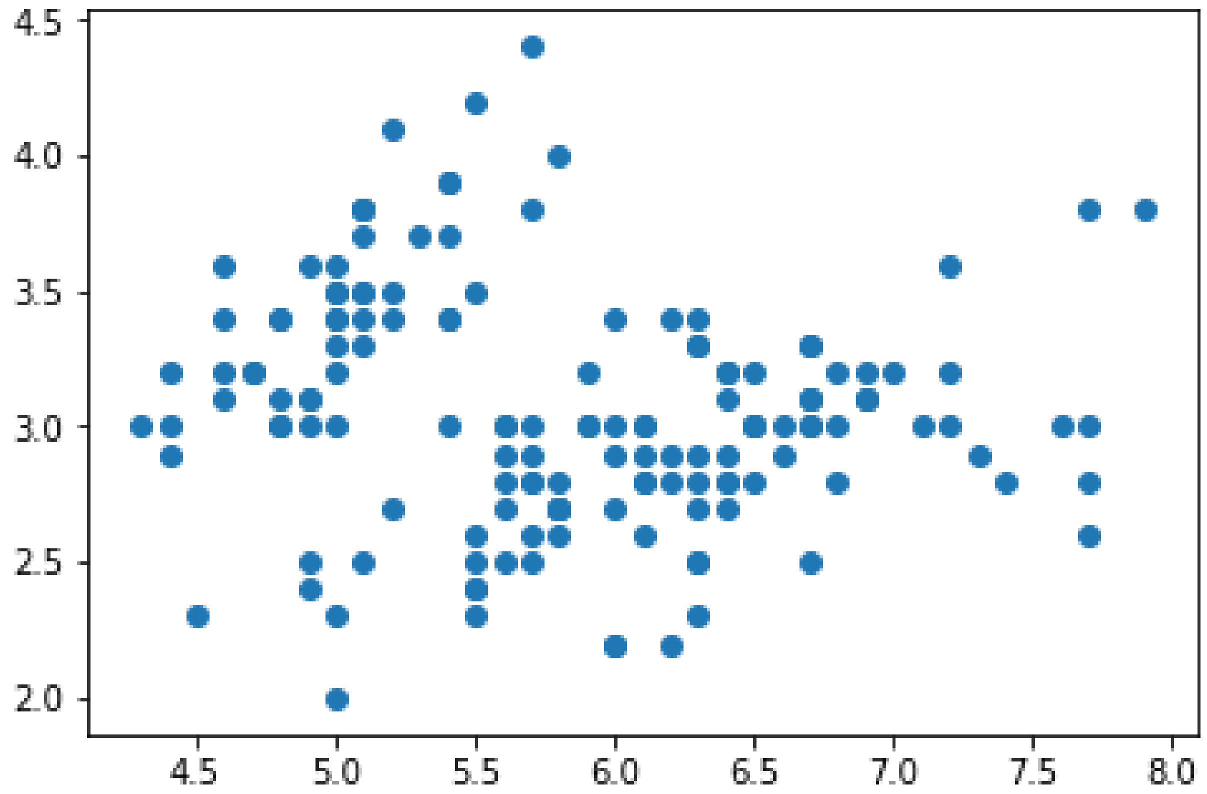
Matplotlib plots

```
import matplotlib.pyplot as plt
plt.hist(iris['sepal_length'])
plt.show()
```



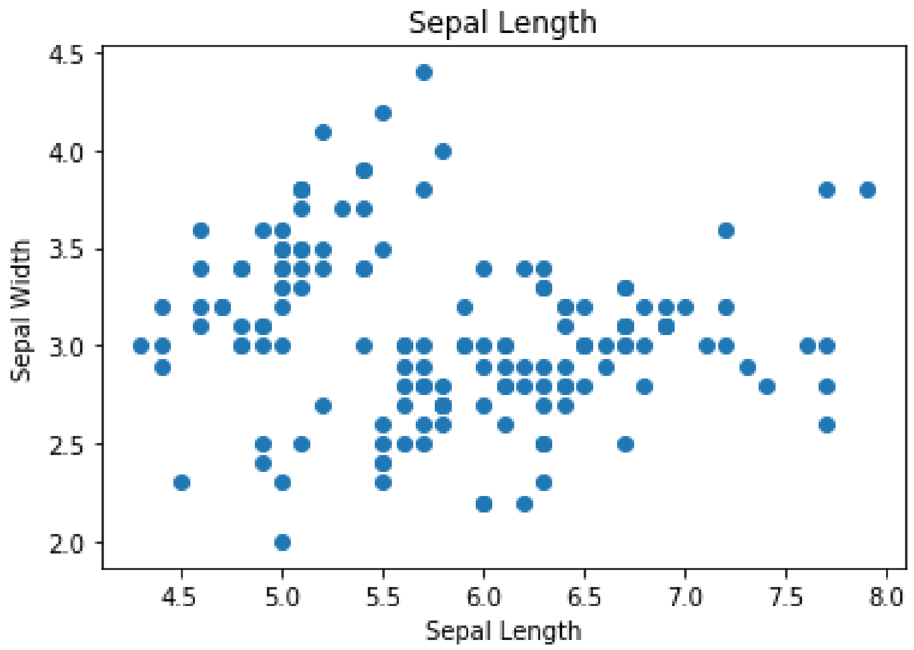
Matplotlib scatter

```
plt.scatter(iris['sepal_length'], iris['sepal_width'])
plt.show()
```



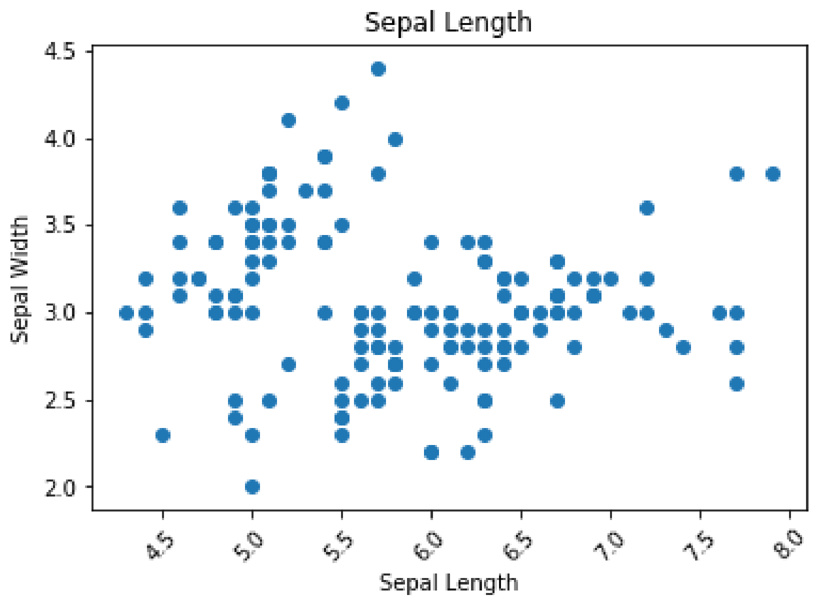
Polishing up the figure

```
fig, ax = plt.subplots()
ax.scatter(iris['sepal_length'], iris['sepal_width'])
ax.set_title('Sepal Length')
ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
plt.show()
```

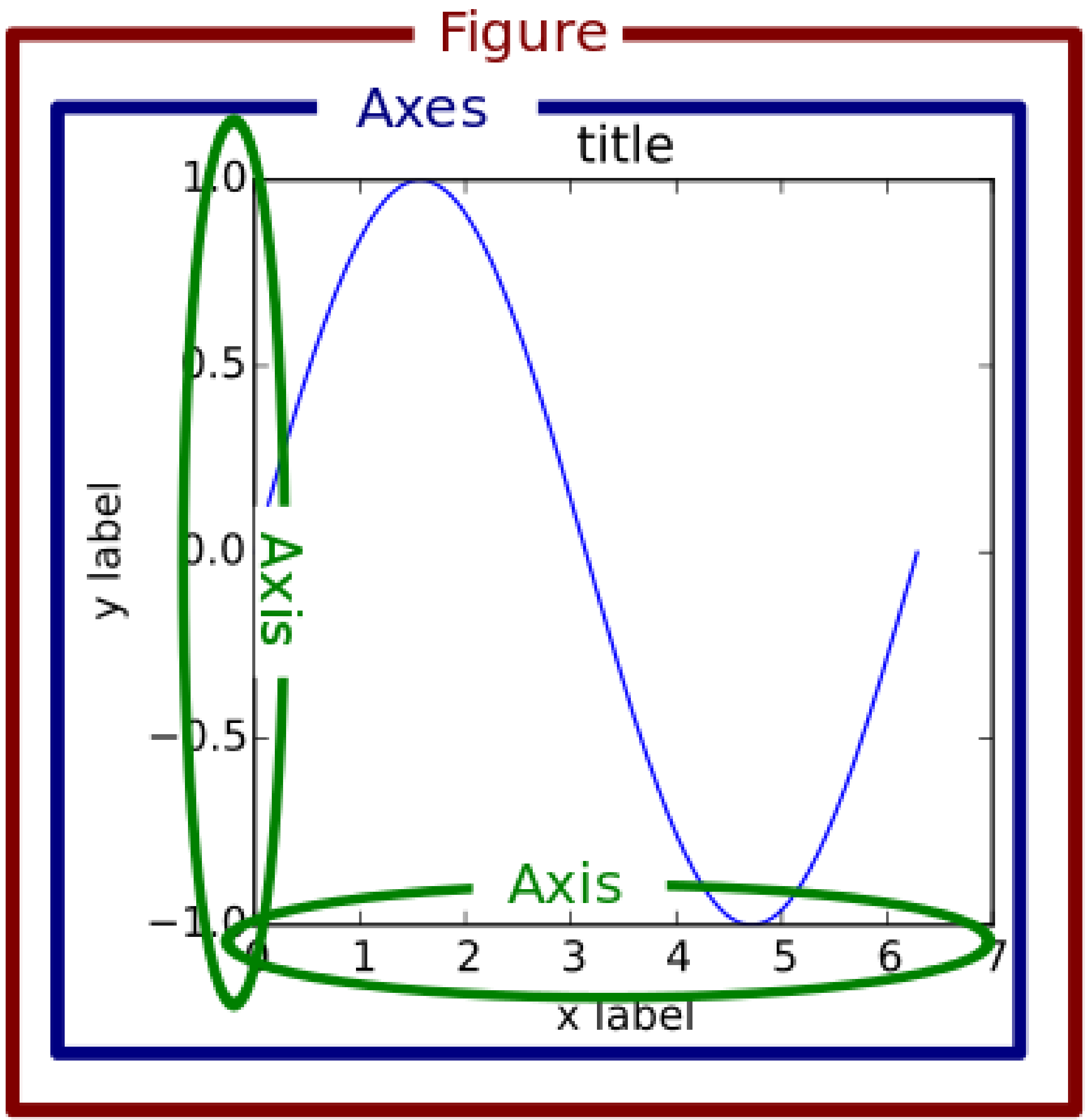


Rotating axis ticks

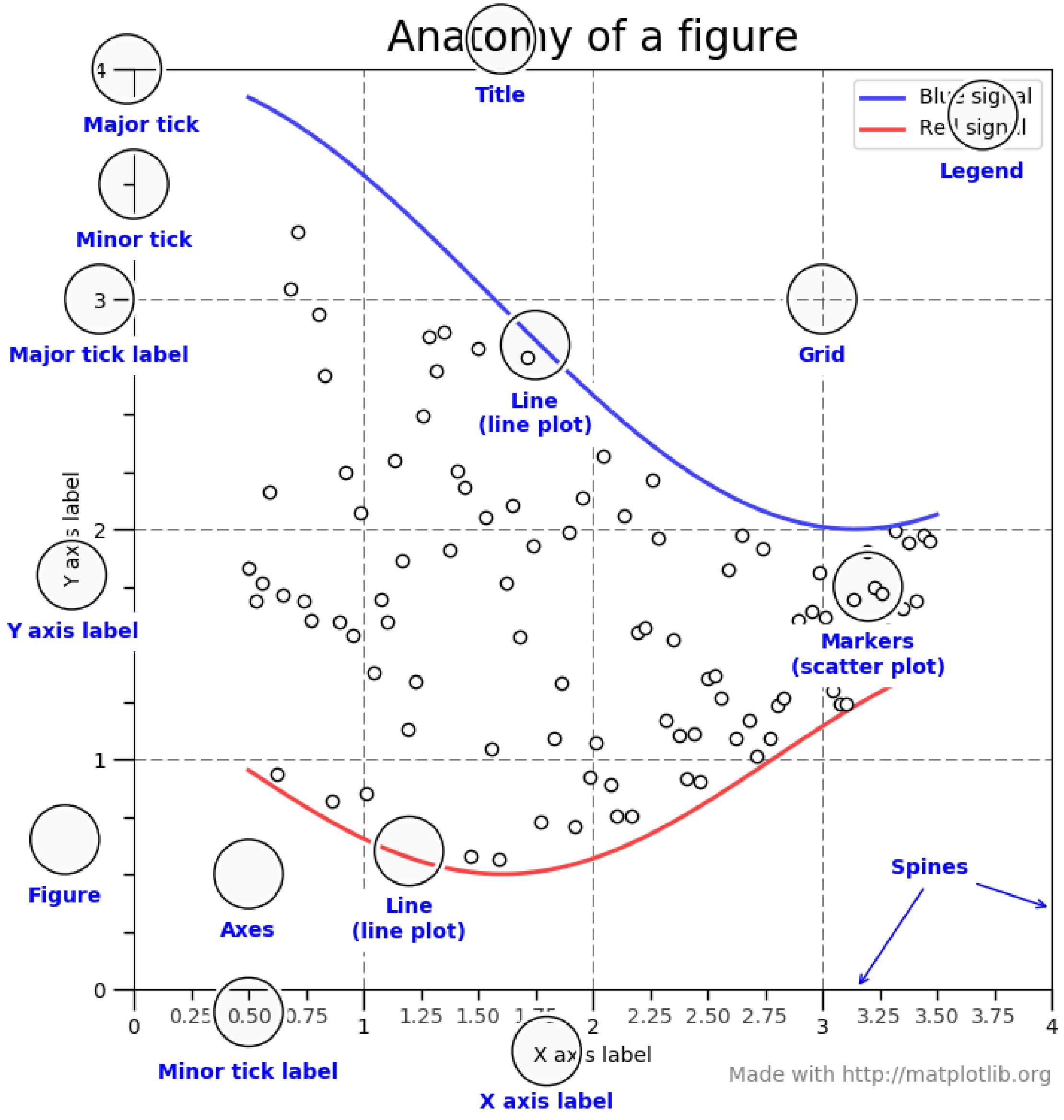
```
fig, ax = plt.subplots()
ax.scatter(iris['sepal_length'], iris['sepal_width'])
ax.set_title('Sepal Length')
ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
plt.xticks(rotation=45) # rotate the x-axis ticks
plt.show()
```



Parts of a matplotlib figure

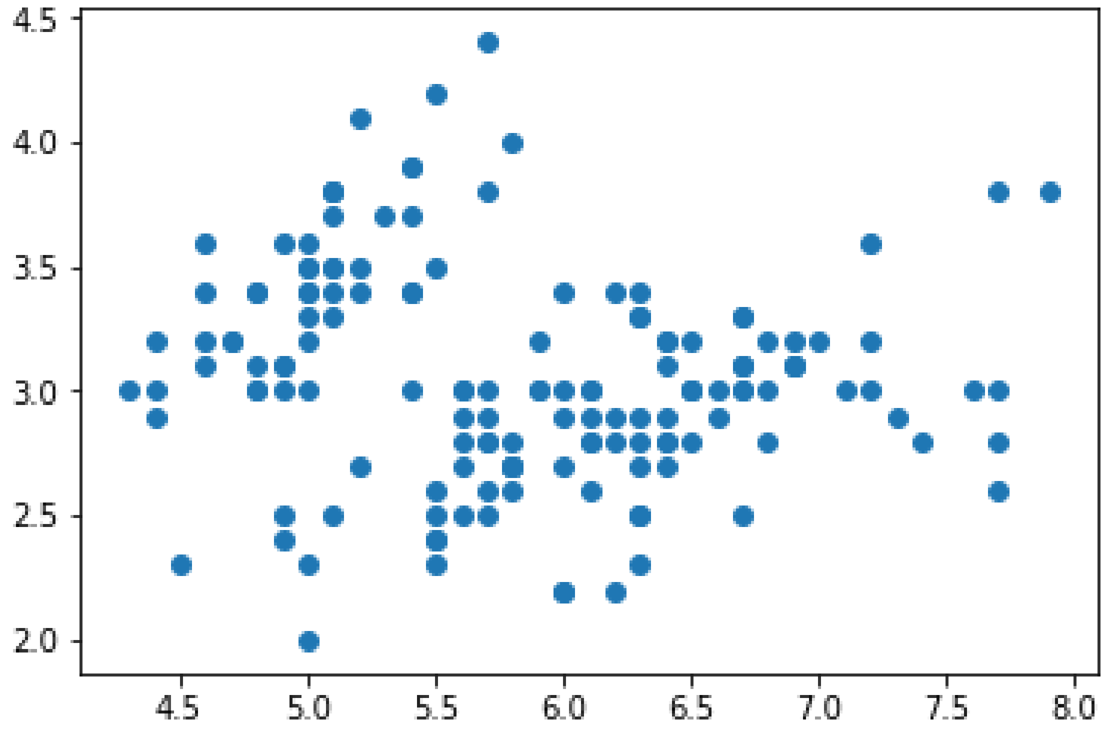


Parts of a matplotlib figure 2



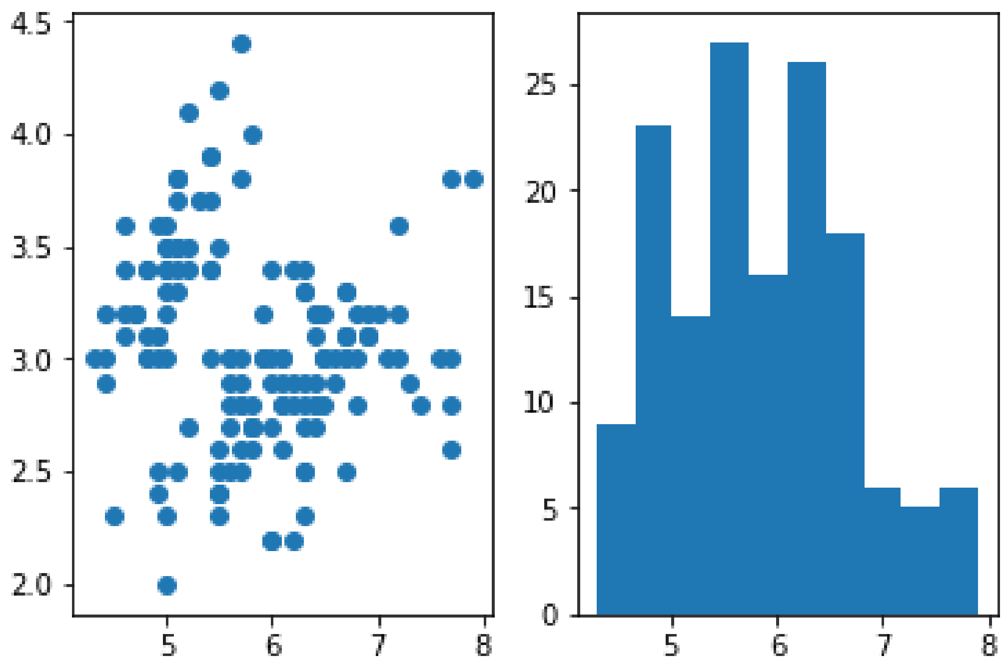
Figures and axes

```
fig, ax = plt.subplots()
ax.scatter(iris['sepal_length'], iris['sepal_width'])
plt.show()
```



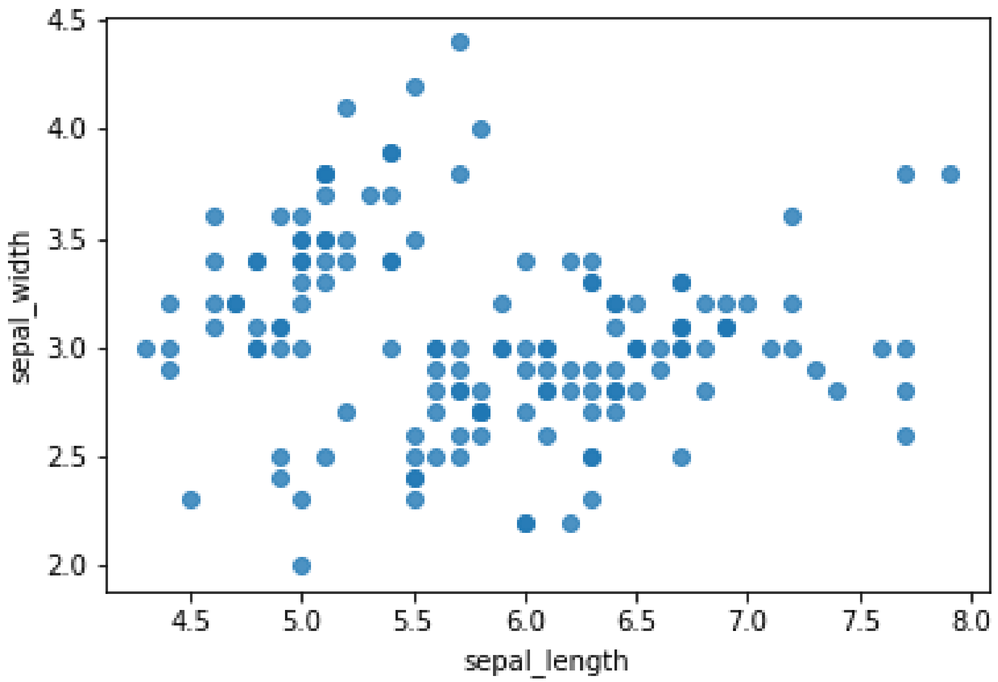
Multiple axes

```
fig, (ax1, ax2) = plt.subplots(1, 2)
ax1.scatter(iris['sepal_length'], iris['sepal_width'])
ax2.hist(iris['sepal_length'])
plt.show()
```



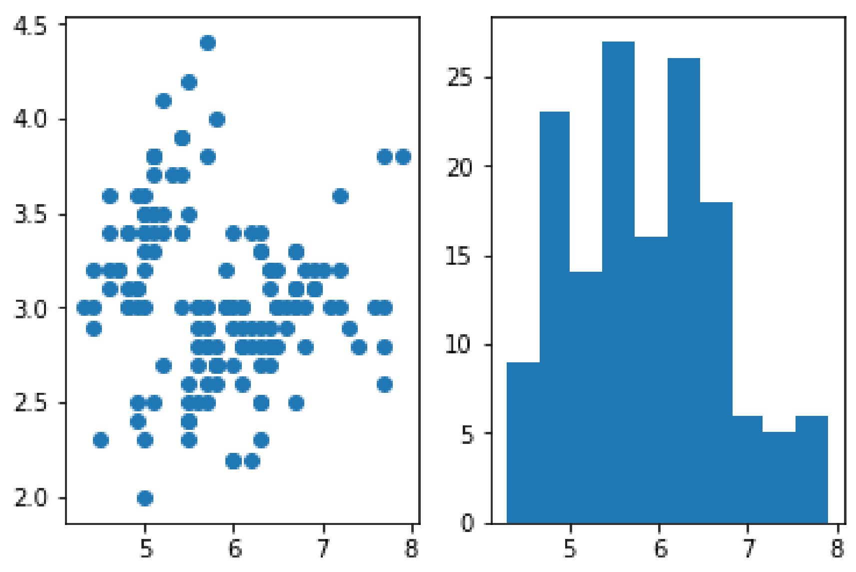
Remember

```
fig, ax = plt.subplots()
sns.regplot(x='sepal_length', y='sepal_width',
            data=iris, fit_reg=False, ax=ax)
plt.show()
```



Clearing the figure

```
fig, (ax1, ax2) = plt.subplots(1, 2)
ax1.scatter(iris['sepal_length'], iris['sepal_width'])
ax2.hist(iris['sepal_length'])
plt.show()
plt.clf()
```



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